

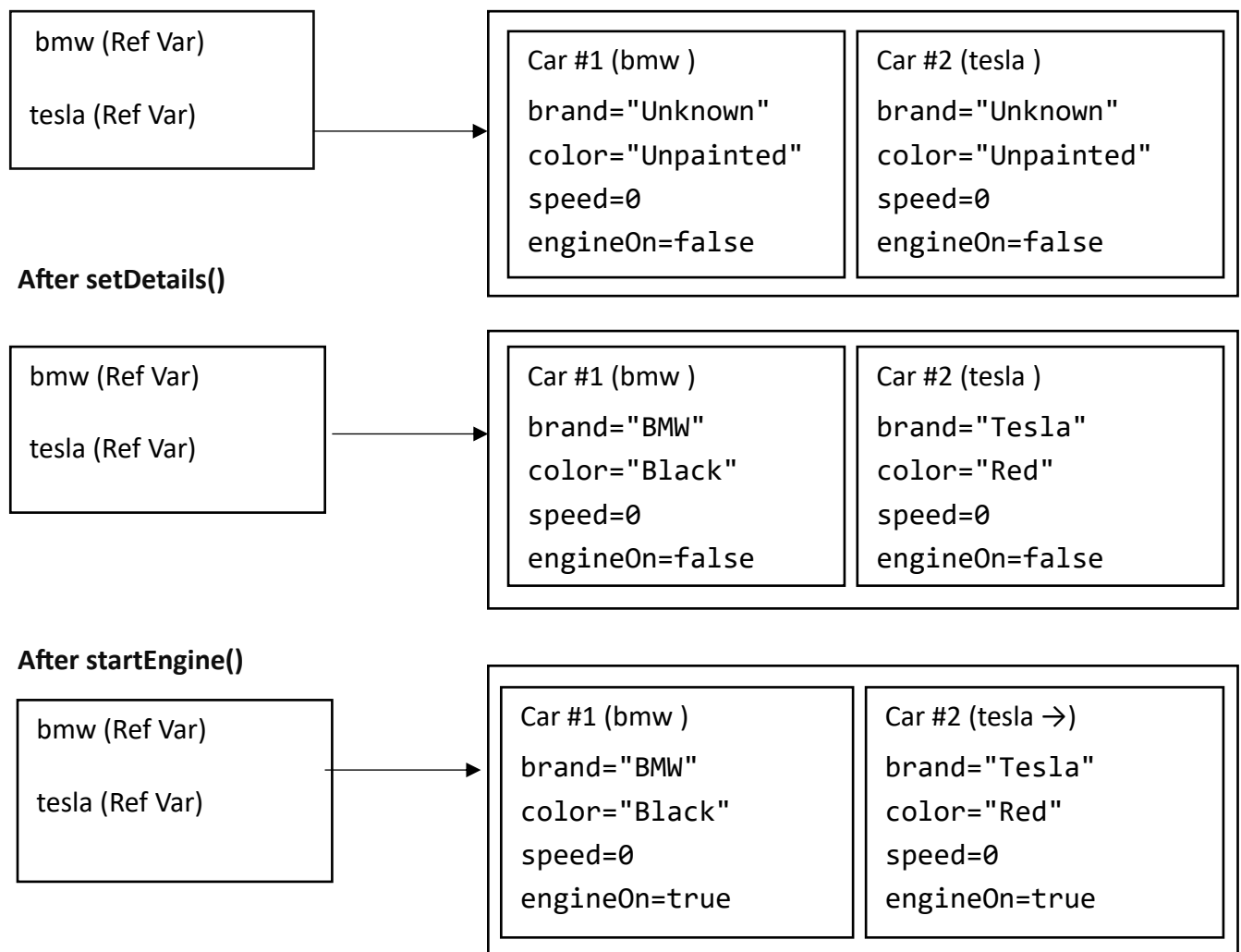
Exercise 2: Draw Stack and Heap in-front of code. Also put necessary comments e.g. instantiation, null construction, reference.

```
public class Main {  
    public static void main(String[] args) {  
  
        // 'Car bmw;' alone (without 'new') would create a reference variable  
        // on the stack holding the value null -- it would point to nothing yet.  
  
        // 'new Car()' allocates a new Car object on the heap and runs the  
        // no-argument constructor.  
        Car bmw = new Car();  
  
        // A second, completely separate object is created here. tesla points  
        // to its own space on the heap, independent of bmw's object.  
        Car tesla = new Car();  
  
        // The dot operator (.) tells Java WHICH object to act on.  
        // Calling methods through the dot operator changes only the fields  
        // that belong to the specific object before the dot.  
        bmw.setDetails("BMW", "Black");  
        tesla.setDetails("Tesla", "Red");  
  
        bmw.startEngine();  
        tesla.startEngine();  
    }  
}
```

```
    bmw.displayState();  
    tesla.displayState();  
  
    bmw.stopEngine();  
    bmw.displayState();  
}  
}
```

STACK

HEAP



After stopEngine()

